# EAB 770 Problem Set 2 Key

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```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(vcdExtra)
```

```
Warning: package 'vcdExtra' was built under R version 4.6.1
```

```
Loading required package: vcd
```

```
Warning: package 'vcd' was built under R version 4.6.1
```

```
Loading required package: grid
Loading required package: gnm
```

```
Warning: package 'gnm' was built under R version 4.6.1
```

```
Registered S3 methods overwritten by 'vcdExtra':
  method      from
  pairs.table vcd
  print.Kappa vcd

Attaching package: 'vcdExtra'

The following objects are masked from 'package:vcd':

    pairs_diagonal_mosaic, woolf_test
```

```
library(vcd)
library(DescTools)
```

```
Warning: package 'DescTools' was built under R version 4.6.1
```

# 1 Problem 1

The table below shows data from a survey of first- and second-generation Filipinos from California, Nevada, and New Jersey about their preferred language to use at home.

```
p1 <- expand.grid(Score = c("All F","More F","Equal","More E","All E"),
                  Generation=c("First","Second"),
                  State = c("Nevada","California","New Jersey"))

p1$Count <- c(17,112,80,25,16,
              8,23,50,79,140,
              13,85,90,31,31,
              5,34,43,123,45,
              6,50,107,64,73,
              10,47,89,52,102)

p1_cc <- xtabs(Count~Generation+Score + State,data=p1)
p1_cc
```

```
, , State = Nevada

          Score
Generation All F More F Equal More E All E
    First     17    112    80     25    16
```

```
    Second     8     23    50     79   140

, , State = California

          Score
Generation All F More F Equal More E All E
    First     13     85    90     31    31
    Second     5     34    43    123    45

, , State = New Jersey

          Score
Generation All F More F Equal More E All E
    First      6     50   107     64    73
    Second    10     47    89     52   102
```

## 1.1 Problem 1a

What is the appropriate extended Mantel-Haenszel statistic to use in testing for a difference in preferred language at home between generations?

> **Solution**
>
> The appropriate MH statistics are the correlation statistic and the mean score statistic.

## 1.2 Problem 1b

Is there sufficient evidence to conclude that there is a difference in preferred language to use at home between the two generations?

> **Solution**

```
CMHtest(p1_cc)
```

```
$`State:Nevada`
Cochran-Mantel-Haenszel Statistics for Generation by Score
    in stratum State:Nevada

                 AltHypothesis  Chisq Df       Prob
cor        Nonzero correlation 174.97  1 6.0709e-40
rmeans  Row mean scores differ 174.97  1 6.0709e-40
```

```
cmeans  Col mean scores differ 192.14  4 1.8428e-40
general    General association 192.14  4 1.8428e-40


$`State:California`
Cochran-Mantel-Haenszel Statistics for Generation by Score
    in stratum State:California

                AltHypothesis  Chisq Df       Prob
cor        Nonzero correlation 57.838  1 2.8468e-14
rmeans  Row mean scores differ 57.838  1 2.8468e-14
cmeans  Col mean scores differ 99.363  4 1.3444e-20
general    General association 99.363  4 1.3444e-20


$`State:New Jersey`
Cochran-Mantel-Haenszel Statistics for Generation by Score
    in stratum State:New Jersey

                AltHypothesis  Chisq Df     Prob
cor        Nonzero correlation 2.1305  1 0.144398
rmeans  Row mean scores differ 2.1305  1 0.144398
cmeans  Col mean scores differ 8.7783  4 0.066887
general    General association 8.7783  4 0.066887
```

At the 0.05 level of significance, there is evidence of a difference in preferred language between two generations for Nevada and California ($p<0.0001$), but not New Jersey ($p=0.144$).

## 1.3 Problem 1c

Is there sufficient evidence to conclude that there is a difference in preferred language of Filipinos to use at home between generations after controlling for location?

**i** Solution

```
CMHtest(p1_cc,
        overall=TRUE)$ALL
```

```
Cochran-Mantel-Haenszel Statistics for Generation by Score
    Overall tests, controlling for all strata
```

```
                AltHypothesis  Chisq Df       Prob
cor        Nonzero correlation 168.35  1 1.7001e-38
rmeans  Row mean scores differ 168.35  1 1.7001e-38
cmeans  Col mean scores differ 196.97  4 1.6865e-41
general    General association 196.97  4 1.6865e-41
```

At the 0.05 level of significance, there is evidence of a difference in preferred language between two generations after controlling for state ($\chi^2_1 = 168.35$,p<0.0001).

## 1.4 Problem 1d

Generation can also be interpreted as ordinal. We can use Somers' Delta or Stuart's tau

### 1.4.1 Nevada

```
SomersDelta(p1_cc[,,1],
            direction="column",
            conf.level = 0.95)
```

```
   somers    lwr.ci    upr.ci
0.6420133 0.5724808 0.7115459
```

```
StuartTauC(p1_cc[,,1],
            conf.level = 0.95)
```

```
     tauc    lwr.ci    upr.ci
0.6367074 0.5670685 0.7063463
```

### 1.4.2 California

```
SomersDelta(p1_cc[,,2],
            direction="column",
            conf.level = 0.95)
```

```
   somers    lwr.ci    upr.ci
0.3968480 0.3054616 0.4882344
```

```
StuartTauC(p1_cc[,,2],
           conf.level = 0.95)
```

```
     tauc    lwr.ci    upr.ci
0.3968480 0.3054616 0.4882344
```

### 1.4.3 New Jersey

```
SomersDelta(p1_cc[,,3],
            direction="column",
            conf.level = 0.95)
```

```
     somers      lwr.ci      upr.ci
 0.07221111 -0.01683079  0.16125301
```

```
StuartTauC(p1_cc[,,3],
           conf.level = 0.95)
```

```
       tauc      lwr.ci      upr.ci
 0.07221111 -0.01683079  0.16125301
```

## 2 Problem 2

The data shown below comes from a pilot study investigating visual perception of a mean shift in a data set. The participants were asked to identify plots with a mean shift among a lineup of "null plots", i.e., plots with no shifts. The researchers then recorded if the participants were able to correctly identify the plot with the mean shift for different shift magnitudes and direction.

```
p2 <- expand.grid(Correct=c("Yes","No"),
                  Magnitude = c("Low","Medium","High"),
                  Direction = c("Positive","Negative")
                  )
p2$Count <- c(4,4,5,1,6,0,3,4,3,3,6,3)

p2_cc <- xtabs(Count~Magnitude + Correct + Direction, data=p2)
p2_cc
```

```
, , Direction = Positive

         Correct
Magnitude Yes No
   Low      4  4
   Medium   5  1
   High     6  0

, , Direction = Negative

         Correct
Magnitude Yes No
   Low      3  4
   Medium   3  3
   High     6  3
```

## 2.1 2a

Provide an estimate and asymptotic 95% confidence interval for the following measures of association between shift magnitude and correct identification for positive and negative shifts:

### 2.1.1 Somers' Delta

Solution

```r
# Positive
SomersDelta(p2_cc[,,1],
            direction="column",
            conf.level = 0.95)
```

```
     somers      lwr.ci      upr.ci
-0.34848485 -0.60804938 -0.08892031
```

```r
# Negative
SomersDelta(p2_cc[,,2],
            direction="column",
            conf.level = 0.95)
```

```
    somers     lwr.ci     upr.ci
-0.1698113 -0.5039340  0.1643114
```

## 2.2 2b

At $\alpha$=0.05, what is your conclusion regarding correlation of the level of shift magnitudes and correct identification?

Solution

```
CMHtest(p2_cc, overall=T)
```

```
$`Direction:Positive`
Cochran-Mantel-Haenszel Statistics for Magnitude by Correct
    in stratum Direction:Positive

                 AltHypothesis  Chisq Df     Prob
cor        Nonzero correlation 4.4976  1 0.033943
rmeans  Row mean scores differ 4.6444  2 0.098055
cmeans  Col mean scores differ 4.4976  1 0.033943
general    General association 4.6444  2 0.098055


$`Direction:Negative`
Cochran-Mantel-Haenszel Statistics for Magnitude by Correct
    in stratum Direction:Negative

                 AltHypothesis   Chisq Df    Prob
cor        Nonzero correlation 0.88707  1 0.34627
rmeans  Row mean scores differ 0.92500  2 0.62971
cmeans  Col mean scores differ 0.88707  1 0.34627
general    General association 0.92500  2 0.62971


$ALL
Cochran-Mantel-Haenszel Statistics for Magnitude by Correct
    Overall tests, controlling for all strata

                 AltHypothesis  Chisq Df    Prob
cor        Nonzero correlation 4.2823  1 0.03851
rmeans  Row mean scores differ 4.3131  2 0.11572
cmeans  Col mean scores differ 4.2823  1 0.03851
general    General association 4.3131  2 0.11572
```

There is strong evidence of a correlation between magnitude and correct identification for

positive shifts ($\chi^2_1 = 4.50$, p=0.03), but no evidence for negative shifts ($\chi^2_1 = 0.89$,p=0.35). Overall, there is strong evidence of a conditional association between magnitude and correct identification after controlling for shift direction ($\chi^2_1 = 4.28$, p=0.04).

## 2.3 2c

Perform an exact test to determine whether the level of shift magnitude is correlated with correct identification.

**Solution**

We can use Fisher's Exact Test to test the marginal association between shift magnitude and identification accuracy.

```
p2_marg <- xtabs(Count~Magnitude+Correct,data=p2)
fisher.test(p2_marg)
```

```
    Fisher's Exact Test for Count Data

data:  p2_marg
p-value = 0.1734
alternative hypothesis: two.sided
```

Note that we cannot do an exact test of conditional association in `R`, but we can do it in `SAS`.

## 2.4 2d

Which results should we use in reporting the relationship between shift magnitude and correct identification: asymptotic or exact? Justify in at most two sentences.

**Solution**

The total number of samples at each strata is greater than 20, which means the asymptotic chi-square distribution assumption holds (correlation statistic sample size requirement). We don't need to perform an exact test even if we have small cell counts.

# 3 Problem 3

350 graduate students and 85 professors were asked about their favorite graph type and programming language. The data is in PS2_3.csv, which can be downloaded on Canvas.

```
p3 <- read.csv("PS2_3.csv")
glimpse(p3)
```

```
Rows: 435
Columns: 4
$ ID      <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18,~
$ Class   <chr> "Professor", "Professor", "Professor", "Professor", "Professor~
$ program <chr> "C++", "Python", "Python", "Python", "Python", "Python", "Stat~
$ graph   <chr> "Other", "Heatmap", "Line", "StackedBar", "Bar", "StackedBar",~
```

## 3.1 3a

What is the appropriate extended Mantel-Haenszel statistic to use in testing for an association between favorite graph type and programming language?

**Solution**

The variables are both nominal, hence the appropriate MH statistic is the general association statistic.

## 3.2 3b

Provide an appropriate measure of association between preferred graph type and programming language.

**Solution**

The variables are both nominal. We should report Lambda or the uncertainty coefficient.

```
p3_cc <- xtabs(~program+graph,data=p3)
Lambda(p3_cc, direction="symmetric",conf.level = 0.95)
```

```
    lambda     lwr.ci     upr.ci
0.01475410 0.00000000 0.05260101
```

```
UncertCoef(p3_cc, direction="symmetric",conf.level = 0.95)
```

```
          uc       lwr.ci       upr.ci
 0.009181104 -0.001078617  0.019440824
```

You can also choose to report the asymmetric values.

## 3.3 3c

Is there evidence of an association between favorite graph type and programming language?

**Solution**

You need to indicate whether we're testing for a marginal association or conditional association.
Marginal association ignores professor classification.

```
CMHtest(p3_cc)
```

```
Cochran-Mantel-Haenszel Statistics for program by graph

                 AltHypothesis     Chisq Df    Prob
cor        Nonzero correlation  0.035842  1 0.84984
rmeans  Row mean scores differ  0.065410  3 0.99564
cmeans  Col mean scores differ  2.608232  5 0.76011
general    General association 12.259688 15 0.65928
```

There is no evidence of a marginal association between programming language and favorite graph type ($ ^2_{15}=12.26, p=0.66).
Conditional association controls for professor classification.

```
p3_cc_3 <- xtabs(~program+graph+Class,data=p3)

CMHtest(p3_cc_3,overall=T)$ALL
```

```
Cochran-Mantel-Haenszel Statistics for program by graph
    Overall tests, controlling for all strata

                 AltHypothesis     Chisq Df    Prob
cor        Nonzero correlation 0.0039985  1 0.94958
```

```
rmeans  Row mean scores differ  0.042299  3 0.99772
cmeans  Col mean scores differ    2.8881  5 0.71723
general    General association    12.506 15 0.64037
```

There is no evidence of a conditional association between programming language and favorite graph type ($ ^2_{15}=12.51, p=0.64).

# 4 Problem 4

Two dentists were tasked with rating the bites of 25 children. The ratings for all the children can be found in PS2_4.csv

```
p4 <- read.csv("PS2_4.csv")
glimpse(p4)
```

```
Rows: 25
Columns: 5
$ ID    <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 1~
$ O1    <chr> "DentistA", "DentistA", "DentistA", "DentistA", "DentistA", "Den~
$ O2    <chr> "DentistB", "DentistB", "DentistB", "DentistB", "DentistB", "Den~
$ Bite1 <chr> "II", "I", "I", "III", "I", "II", "I", "I", "III", "II", "II", "~
$ Bite2 <chr> "III", "I", "II", "I", "III", "II", "III", "I", "III", "III", "I~
```

## 4.1 4a

Calculate the estimate and 95% confidence interval of the Fleiss-Cohen weighted kappa.

**Solution**

```
p4_cc <- xtabs(~Bite1+Bite2,data=p4)
p4_cc
```

```
     Bite2
Bite1 I II III
  I   5  3   2
  II  0  5   2
  III 3  0   5
```

```
Kappa(p4_cc, weight="Fleiss-Cohen")
```

```
             value    ASE     z
Unweighted 0.4005 0.1470 2.725
Weighted   0.2890 0.2104 1.373
```

### 4.2 4b

Describe the strength of rating agreement between the two dentists.

**Solution**

The agreement is weak based on the weighted $\kappa$, which means there might be a lot of high discordance in the ratings.